

KIRTAN RAJESH

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Education

Vellore Institute of Technology, Chennai

2021 – 2025

B.Tech. Computer Science and Engineering (Specialization: AI & Robotics)

CGPA: 8.72/10.0

• **Relevant Coursework:** Data Structures & Algorithms, OOP, DBMS, Software Engineering, Distributed Systems

Work Experience

BETTROI

November 2025 – January 2026

AI Solutions Engineer (Remote)

United Arab Emirates

- Designed and implemented scalable REST API services using Python/Flask; architected Retrieval Augmented Generation (RAG) pipelines with vector databases, enabling document Q&A at sub-second latency across 10K+ document embeddings
- Built modular, microservice-style LLM application architecture integrating multiple AI APIs (OpenAI, Groq); implemented FAISS-based semantic search and embedding pipelines, improving context retrieval accuracy by 30% over keyword-based approaches

Hucenrotia Lab, National Yang Ming Chiao Tung University

August 2025 – October 2025

Research Intern – Robotics & AI

NYCU Hsinchu, Taiwan

- Developed high-performance control software in C++ and Java for autonomous robot simulation, achieving 77.8% ball interception accuracy; applied SDLC best practices with modular code architecture and clean component interfaces enabling parallel team development
- Implemented Extended Kalman Filter for real-time multi-sensor data fusion (camera and position data); engineered PID-controlled trajectory planning system and inverse kinematics solver, reducing motion planning latency by 40%

Caliber Interconnect

June 2024 – July 2024

Software Developer Intern

Bengaluru, Karnataka

- Built real-time data visualization and control dashboard using React and WebSocket protocol (ROSBridge), enabling low-latency bidirectional communication between a React frontend and distributed robot simulation backend
- Improved frontend performance and scalability through component refactoring, code-splitting, and efficient state management using React hooks, reducing unnecessary re-renders and improving load time by 35%

DelaPlex Ltd.

August 2023 – September 2023

Software Engineer Intern

Nagpur, Maharashtra

- Built end-to-end recommendation system with Java/Flask REST API backend and MySQL relational database; designed normalized database schema with optimized SQL queries and indexing strategies, reducing average query execution time by 30%
- Engineered ETL data processing pipelines to ingest, validate, and transform user interaction datasets; implemented robust data quality checks and error handling ensuring 99%+ data integrity across pipeline workflows

Software Development Projects

Pulse of People – AI-Powered Sentiment Analytics Platform | *React, Flask, MySQL, Docker*

- Developed full-stack political analytics application processing 50K+ sentiment records; architected MySQL relational database with efficient indexing, optimized SQL queries, and server-side caching, achieving 40% reduction in API response times
- Implemented containerized microservice deployment using Docker; designed RESTful API layer with clear separation of concerns between data processing, business logic, and presentation tiers

ScanittAI – Intelligent Document Management System | *Flask, React, OpenCV, SQL, Groq AI*

- Built production-grade document processing system with Flask REST API, OpenCV image preprocessing pipeline, and Tesseract OCR; designed relational database schema for document metadata with full-text search capabilities

Publications & Research

[**First Author**] Deep Reinforcement Learning for Urban Air Quality Management – IEEE Access, 2025 (IF: 3.9). Designed PPO-based RL framework; conducted comprehensive data analysis on 3+ years of datasets with 1M+ data points; achieved 26% AQI improvement through systematic experimentation.

[**First Author**] MSHAN: Multi-Scale Hybrid Attention Network for ECG Arrhythmia Detection – IEEE ICCNT, 2025. Developed hybrid deep learning architecture combining CNNs and attention mechanisms; achieved 68.78% accuracy with SHAP interpretability analysis.

[**Co-Author**] Resilient Solar Microgrid Management Using Multi-Agent RL – Results in Engineering, Elsevier, 2025. Contributed to multi-agent RL system design using DQN.

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript

Frameworks & APIs: Spring Boot (familiar), React, Flask, Node.js, REST APIs, HTML/CSS

Databases: MySQL, PostgreSQL, MongoDB, Redis; relational schema design & query optimization

Infrastructure & Tools: Docker, Git, CI/CD (GitHub Actions), Linux/Unix, Postman

AI/ML: LangChain, OpenAI API, TensorFlow, Scikit-learn, FAISS, RAG systems

Leadership & Achievements

Google Developers Student Club – Management Team Lead: Organized 3 national hackathons with 200+ participants

LeetCode Profile: Solved 300+ problems focusing on algorithms, data structures and system design

State Powerlifting Champion: #1 VIT Chennai, #12 Tamil Nadu (2023–24)

Certifications

Generative AI Fundamentals – Databricks

Supervised Machine Learning: Regression and Classification – Coursera

The Complete Full-Stack Web Development Bootcamp – Udemy

Recommender Systems and Deep Learning in Python – Udemy

TensorFlow Developer Certificate – Udemy